9/29/25 | ECC /HW4 | Walker, Nathan

4.2 In a ring with identity (that is, multiplicative identity), denote this identity as 1. Prove:

(a) The multiplicative identity is unique.

(b) If an element a has both a right inverse b (i.e., an element b such that $ab = 1$) and a left inverse c (i.e., an element c such that $ca = 1$), then $b = c$. In this case, the element a is said to have an inverse (denoted by a^{-1}). Show that the inverse of an element a, when it exists, is unique.

(c) If a has a multiplicative inverse a^{-1}, then $(a^{-1})^{-1} = a$.

(d) The set of units of a ring forms a group under multiplication. (Recall that a unit of a ring is an element that has a multiplicative inverse.)

(e) If $c = ab$ and c is a unit, then a has a right inverse and b has a left inverse.

(f) In a ring, a nonzero element a such that $ax = 0$ for $x \neq 0$ is said to be a zero divisor. Show that if a has an inverse, then a is not a zero divisor.

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a) $R = \langle \Omega, +, \cdot \rangle$

e_1 is an identity, e_2 is also an identity

$e_1 \cdot a = a \qquad e_2 \cdot a = a$

$(e_1 \cdot a = e_2 \cdot a)\, a^{-1}$

$e_1 \cdot e_1 = e_2 \cdot e_2$

$e_1 = e_2$

$\therefore$ unique

b) Suppose two unique inverses b & c

$a \cdot b = e \qquad c \cdot a = e$

$a(a \cdot b = c \cdot a)$

$a \cdot a \cdot b = a \cdot (c \cdot a) = a \cdot e = a$

$a(a \cdot b) = a$

$a \cdot e = a = a$

$\therefore$ unique

c) $(a^{-1} \cdot a = e)\, a \Rightarrow (a^{-1} a) a = (e \cdot a)$

$e \cdot a = a$

$a = a$

d) $ab = b \cdot a = 1 \qquad b = a^{-1}$ & $a = b^{-1}$

I) if $u, v \in U(R)$ check uv, unit if $(uv)(v^{-1}u^{-1}) = 1$

$u(vv^{-1})u^{-1} = u(1)u^{-1} = uu^{-1} = 1$

II) Associativity inherited from ring

III) 1 is the Identity

IV) every u in $\in U(R)$ has an inverse by definition

$\therefore$ its a group

e) $c = ab \qquad cc^{-1} = 1$

$c \cdot c^{-1} = a(bc^{-1}) = a a^{-1}$

$c^{-1} \cdot c = (c^{-1}a)b = b^{-1}b$

f) $(a \cdot a^{-1} = e)x$

$a \cdot a^{-1}x = x$

$a \cdot a^{-1}x - x = 0$

$(a a^{-1} - 1)x = 0$ ($e = 1$)

$(1 - 1)x = 0$

$0 \cdot x = 0 \quad \therefore$ not a divisor

4.3 Construct the ring $R_4 = GF(2)[x]/(x^4 + 1)$. That is, construct the addition and multiplication tables for the ring. Is R_4 a field?

$\Omega = 1, x, 1+x, x^2, 1+x^2, x+x^2, 1+x+x^2, x^3, x^3+1, x^3+x, x^3+x+1, x^3+x^2$
$x^3+x^2+1, x^3+x^2+x, x^3+x^2+x+1$

represent as binary and then as decimal

+	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1	0	3	2	5	4	7	6	9	8	11	10	13	12	15	14
2	2	3	0	1	6	7	4	5	10	11	8	9	14	15	12	13
3	3	2	1	0	7	6	5	4	11	10	9	8	15	14	13	12
4	4	5	6	7	0	1	2	3	12	13	14	15	8	9	10	11
5	5	4	7	6	1	0	3	2	13	12	15	14	9	13	11	15
6	6	7	4	5	2	3	0	1	14	15	12	13	10	11	8	9
7	7	6	5	4	3	2	1	0	15	14	13	12	11	10	9	8
8	8	9	10	11	12	13	14	15	0	1	2	3	4	5	6	7
9	9	8	11	10	13	12	15	14	1	0	3	2	5	4	7	6
10	10	11	8	9	14	15	12	13	2	3	0	1	6	7	4	5
11	11	10	9	8	15	14	13	12	3	2	1	0	7	6	5	4
12	12	13	14	15	8	9	10	11	4	5	6	7	0	1	2	3
13	13	12	15	14	9	13	11	10	5	4	7	6	1	0	3	2
14	14	15	12	13	10	11	8	9	6	7	4	5	2	3	0	1
15	15	14	13	12	11	15	9	8	7	6	5	4	3	2	1	0

*	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
2	0	2	4	6	8	10	12	14	1	3	5	7	9	11	13	15
3	0	3	6	5	12	15	10	9	9	10	15	12	5	6	3	0
4	0	4	8	12	1	5	9	13	2	6	10	14	3	7	11	15
5	0	5	10	15	5	0	15	10	10	15	0	5	15	10	5	0
6	0	6	12	10	9	15	5	3	3	5	15	9	10	12	6	0
7	0	7	14	9	13	10	3	4	11	12	5	2	6	1	8	15
8	0	8	1	9	2	10	3	11	4	12	5	13	6	14	7	15
9	0	9	3	10	6	15	5	12	12	5	15	6	10	3	9	0
10	0	10	5	15	10	0	15	5	5	15	0	10	15	5	10	0
11	0	11	7	12	14	5	9	2	13	6	10	1	3	8	4	15
12	0	12	9	5	3	15	10	6	6	10	15	3	5	9	12	0
13	0	13	11	6	7	10	12	1	14	3	5	8	9	4	2	15
14	0	14	13	3	11	5	6	8	7	9	10	4	12	2	1	15
15	0	15	15	0	15	0	0	15	15	0	0	15	0	15	15	0

5 has no inverse so not a field

4.4 Let R be a commutative ring and let $a \in R$. Let $I = \{b \in R : ab = 0\}$. Show that I is an ideal of R.

I) $a \cdot 0 = 0 \quad 0 \in I$

II) $b_1 \cdot b_2 \in I \quad a(b_1 + b_2) \Rightarrow ab_1 + ab_2 = 0 + 0 = 0$

$b_1 + b_2 \in I$

III) $b \in I \quad ab = 0$

$a \cdot b = -ab = -(ab) = 0$

IV) $r \in R$ & $b \in I$

$a(rb) = rab = r \cdot 0 = 0 \quad$ so $rb \in I$

4.5 An element a of a ring R is nilpotent if $a^n = 0$ for some positive integer n. Show that the set of all nilpotent elements in a commutative ring R is an ideal.

if $a^n = 0 \quad$ if $n > 0$

an ideal has property of $a \in I \quad b \notin I \quad a \cdot b \in I$

$a^n = 0 \in I \quad 0 \in I$

$\therefore b \cdot a^n = 0 \quad \forall n > 0 \Rightarrow b \cdot a^n \in I$

adding any a^n w/ $a^m \quad n, m > 0$

$a^n + a^m = 0 + 0 = 0 \in I$

subgroup under addition

4.7 Show that in the ring Z_{15} the polynomial $p(x) = x^2 - 1$ has more than two zeros. In a field there would be only two zeros. What may be lacking in a ring that leads to "too many" zeros?

$x^2 - 1 = 0$

$(x-1)(x+1) = 0$

two roots are 1 and 14

but if $x^2 \bmod 15 = 1$ then another root

take $x = 4 \quad 4^2 = 16 \bmod 15 = 1$

$\therefore 4^2 - 1 = 0$

rules on multiplication is relaxed so this possible

4.9 For the (15, 11) binary Hamming code with generator $g(x) = x^4 + x + 1$:

(a) Determine the parity check polynomial $h(x)$.

(b) Determine the generator matrix G and the parity check matrix H for this code in nonsystematic form.

(c) Determine the generator matrix G and the parity check matrix H for this code in systematic form.

(d) Let $m(x) = x + x^2 + x^3$. Determine the code polynomial $c(x) = g(x)m(x)$.

(e) Let $m(x) = x + x^2 + x^3$. Determine the systematic code polynomial $c(x) = x^{n-k}m(x) + R_{g(x)}[x^{n-k}m(x)]$, where $R_{g(x)}[\ ]$ computes the remainder after division by $g(x)$.

(f) For the codeword $c(x) = 1 + x + x^3 + x^4 + x^5 + x^9 + x^{10} + x^{11} + x^{13}$, determine the message if nonsystematic encoding is employed.

(g) For the codeword $c(x) = 1 + x + x^3 + x^4 + x^5 + x^9 + x^{10} + x^{11} + x^{13}$, determine the message if systematic encoding is employed.

(h) Let $r(x) = x^{14} + x^{10} + x^5 + x^3$. Determine the syndrome for $r(x)$.

(i) Draw the systematic encoder circuit for this code using the $g(x)$ feedback polynomial.

(j) Draw the decoder circuit for this circuit with $r(x)$ input on the left of the syndrome register. Determine in particular the error pattern detection circuit.

(k) Draw the decoder circuit for this circuit with $r(x)$ input on the right of the syndrome register. Determine in particular the error pattern detection circuit.

(l) Let $r(x) = x^{13} + x^{10} + x^9 + x^5 + x^2 + 1$. Trace the execution of the Meggitt decoder with the input on the left, analogous to Table 4.5.

(m) Let $r(x) = x^{13} + x^{10} + x^9 + x^5 + x^2 + 1$. Trace the execution of the Meggitt decoder with the input on the right, analogous to Table 4.6.

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a) $n = 15$ $\quad h(x) = \dfrac{x^{15}+1}{g(x)} =$

a)-h) Screenshot

```
hx =
     1     0     0     1     1     0     1     0     1     1     1     1

G =
     1     0     0     1     1     0     0     0     0     0     0     0     0     0     0
     0     1     0     0     1     1     0     0     0     0     0     0     0     0     0
     0     0     1     0     0     1     1     0     0     0     0     0     0     0     0
     0     0     0     1     0     0     1     1     0     0     0     0     0     0     0
     0     0     0     0     1     0     0     1     1     0     0     0     0     0     0
     0     0     0     0     0     1     0     0     1     1     0     0     0     0     0
     0     0     0     0     0     0     1     0     0     1     1     0     0     0     0
     0     0     0     0     0     0     0     1     0     0     1     1     0     0     0
     0     0     0     0     0     0     0     0     1     0     0     1     1     0     0
     0     0     0     0     0     0     0     0     0     1     0     0     1     1     0
     0     0     0     0     0     0     0     0     0     0     1     0     0     1     1

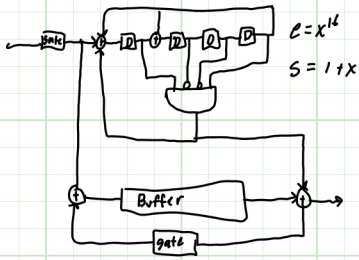
H =
     1     0     0     1     1     0     1     0     1     1     1     1     0     0     0
     0     1     0     0     1     1     0     1     0     1     1     1     1     0     0
     0     0     1     0     0     1     1     0     1     0     1     1     1     1     0
     0     0     0     1     0     0     1     1     0     1     0     1     1     1     1
```

```
Gsys =
     1     0     0     0     0     0     0     0     0     0     0     0     1     0     0     1
     0     1     0     0     0     0     0     0     0     0     0     0     1     1     0     1
     0     0     1     0     0     0     0     0     0     0     0     0     1     1     1     1
     0     0     0     1     0     0     0     0     0     0     0     0     1     1     1     0
     0     0     0     0     1     0     0     0     0     0     0     0     0     1     1     1
     0     0     0     0     0     1     0     0     0     0     0     0     1     0     1     0
     0     0     0     0     0     0     1     0     0     0     0     0     0     1     0     1
     0     0     0     0     0     0     0     1     0     0     0     0     1     0     1     1
     0     0     0     0     0     0     0     0     1     0     0     0     1     1     0     0
     0     0     0     0     0     0     0     0     0     1     0     0     0     1     1     0
     0     0     0     0     0     0     0     0     0     0     1     0     0     1     1     1
H =
     1     0     0     0     1     1     1     1     0     1     0     1     1     0     0
     0     1     0     0     0     1     1     1     1     0     1     0     1     1     0
     0     0     1     0     0     0     1     1     1     1     0     1     0     1     1
     0     0     0     1     1     1     1     0     1     0     1     1     0     0     1
cx1 =
     1     1     1     1     0     0     1     0     0     0     0     0     0     0     0     0
cx2 =
     1     1     1     0     0     0     0     0     0     0     0     0     1     0     1     1
mxf =
     0     1     0     1     0     0     1     1     0     0     1
mxg =
     0     1     0     1     1     1     0     0     0     1     1
synh =
     1     0     0     1     0     0     1     1     0     0     0
```

i)



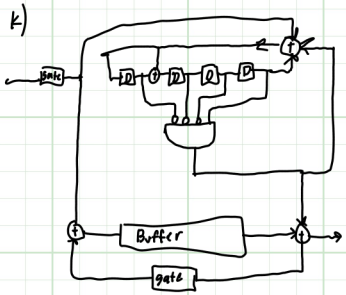
j)



$e = x^{14}$

$s = 1 + x^3$

k)



error: x^{18}

$s = x^3$

l)

step	input	syndrome (left)	buffer (left)	syndrome (right)	buffer (right)
1	0	0000	0000000000000000	0000	0
2	1	1000	100 — 0	1100	10 — 0
3	0	0100	010 — 0	0110	010 — 0
4	0	0010	0010 — 0	0011	0010 — 0
5	1	1001	10010 — 0	1001	10010 — 0
6	1	1100	11001 0 — 0	1100	11001 0 — 0
7	0	0110	01100 1 0 — 0	0110	01100 1 0 — 0
8	0	0011	00110 0 1 0 — 0	0011	00110 0 1 0 — 0
9	0	0001	00011 0 0 1 0 — 0	1001	00011 0 0 1 0 — 0
10	1	1000	10001 1 0 0 1 0 — 0	0100	10001 1 0 0 1 0 — 0
11	0	0100	01000 1 1 0 0 1 0 0	0010	01000 1 1 0 0 1 0 0
12	0	0010	00100 0 1 1 0 0 1 0 0 0	0001	00100 0 1 1 0 0 1 0 0 0
13	1	1001	10010 0 0 1 1 0 0 1 0 0 0	1100	10010 0 0 1 1 0 0 1 0 0 0
14	0	1100	01001 0 0 0 1 1 0 0 1 0 0 0	0110	01001 0 0 0 1 1 0 0 1 0 0 0
15	1	1110	10100 1 0 0 0 1 1 0 0 1 0 0 0	0111	10100 1 0 0 0 1 1 0 0 1 0 0 0
16	0	0111	01010 0 1 0 0 0 1 1 0 0 1 0 0 0	0111	01010 0 1 0 0 0 1 1 0 0 1 0 0 0
17	0	0011	00101 0 0 1 0 0 0 1 1 0 0 1 0 0 0	1110	00101 0 0 1 0 0 0 1 1 0 0 1 0 0 0
18	0	1101	01110 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	1111	01110 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
19	0	1010	10111 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	1011	10111 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
20	0	0101	01011 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	1001	01011 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
21	0	1110	00101 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	0100	00101 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
22	0	0111	00010 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	0010	00010 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
23	0	1111	00001 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	0001	00001 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
24	0	1011	00000 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	1000	00000 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
25	0	1001	00000 0 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	0000	00000 0 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
26	0	0000	00000 0 0 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	0000	00000 0 0 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
27	0	0000	00000 0 0 0 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	0000	00000 0 0 0 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
28	0	0000	00000 0 0 0 0 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	0000	00000 0 0 0 0 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
29	0	0000	00000 0 0 0 0 0 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	0000	00000 0 0 0 0 0 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0
30	0	0000	00000 0 0 0 0 0 0 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0	0000	00000 0 0 0 0 0 0 1 0 1 0 1 0 0 0 0 1 0 0 0 1 1 0 0 1 0 0 0

4.10 Let $f(x)$ be a polynomial of degree m in F[x], where F is a field. Show that if a is a root of $f(x)$ (so that $f(a) = 0$), then $(x - a) \mid f(x)$. *Hint:* Use the division algorithm. Inductively, show that $f(x)$ has at most m roots in F.

$f(x) = q(x)(x-a) + r(x)$ $\quad r(x)$ is a constant

$f(a) = q(a)(a-a) + r(a)$

$f(a) = 0 + r(a)$ $\quad r(a) = 0$ so $f(a) = 0$

also $r(x) = 0$

So $q(x)$ has $m-1$ degree $\therefore$ $f(x)$ has degree m

4.14 [272] Let $g(x)$ be the generator for a binary (n, k) cyclic code C. The reciprocal of $g(x)$ is defined as

$$g^*(x) = x^{n-k} g(1/x).$$

(In this context, "reciprocal" does *not* mean multiplicative inverse.)

(a) As a particular example, let $g(x) = 1 + x^2 + x^4 + x^6 + x^7 + x^{10}$. Determine $g^*(x)$. The following subproblems deal with arbitrary cyclic code generator polynomials $g(x)$.

(b) Show that $g^*(x)$ also generates an (n, k) cyclic code.

(c) Let C^* be the code generated by $g^*(x)$. Show that C and C^* have the same weight distribution.

(d) Suppose C has the property that whenever $c(x) = c_0 + c_1 x + c_{n-1} x^{n-1}$ is a codeword, so is its reciprocal $c^*(x) = c_{n-1} + c_{n-2} x + \cdots + c_0 x^{n-1}$. Show that $g(x) = g^*(x)$. Such a code is said to be a *reversible* cyclic code.

a) $g(1/x) = 1 + x^{-2} + x^{-4} + x^{-6} + x^{-7} + x^{-10}$

$n-k = 10$

$x^{10} g(1/x) = x^{10} + x^8 + x^6 + x^4 + x^3 + 1$

b) $g(x) \mid x^n - 1 \Rightarrow (h(x) g(x) = x^n - 1)^*$

$h^*(x) g^*(x) = x^n - 1$

So $g^*(x)$ is also a cyclic code

c) define $R(c(x)) := x^{n-1} c(1/x)$

$c(x) = c_0 + c_1 x + \ldots + c_{n-1} x^{n-1}$ $\quad x^{n-1} c(1/x) = c_{n-1} + c_{n-2} x + \ldots + c_1 x^{n-2} + c_0 x^{n-1}$

$x^{n-1} m(1/x)\, g(1/x) = x^{k-n+n-1}\, m(1/x)\, g(1/x) = \underbrace{x^{n-1} m(1/x)}_{\text{reverses message}}\, g^*(x)$

R maps C to C^* $\quad m^*(x)\, g^*(x)$

$\therefore$ same wt distribution

d) $c(x) \in C$ $\quad c^*(x) \in C$

$[m(x) g(x)]^* = m^*(x)\, g^*(x) = m(x)\, g^*(x) = c(x)$

4.15 [272] Let $g(x)$ be the generator polynomial of a binary cyclic code of length n.

(a) Show that if $g(x)$ has $x+1$ as a factor, then the code contains no codevectors of odd weight.

(b) Show that if $x+1$ is *not* a factor of $g(x)$, then the code contains the all-one codeword.
Hint: The following is true for any ring $F[x]$:

$$\boxed{1-x^n=(1-x)(1+x+x^2+\cdots+x^{n-1}).}$$

(c) Show that the code has minimum weight at least 3 if n is the smallest integer such that $g(x)$ divides x^n-1.

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a)

$g(x) = (1+x)a(x)$

$c(x) = m(x)(1+x)a(x)$

$c(1) = m(1)(1+1)^0 a(1) = 0 \mod 2$

$\therefore$ even

b) $g(x)h(x) = x^n+1 = (x+1)(1+x+x^2+\ldots+x^{n-1})$

must be a factor of $h(x)$

$m(x) = \frac{h(x)}{(x+1)} \Rightarrow m(x)g(x) = \frac{g(x)h(x)}{(x+1)} = \frac{x^n+1}{x+1} = 1+x+x^2+\ldots+x^{n-1}$

All one codeword

c) wt of 1 : $c(x) = x^i$ $g(x)$ doesn't divide

wt of 2 : $c(x) = x^i + x^j = x^i(1+x^{j-i})$

$j-i < n$

$c(x) = g(x)m(x)$

$g(x)$ cannot divide $j-i<n$ $\therefore$

wt of 2 cannot exist

4.43 Given a sequence $y(x)$ produced by dividing by the reciprocal polynomial of $g(x)$,

$$y(x) = \frac{d(x)}{g^*(x)},$$

determine what $d(x)$ should be to obtain the given $y(x)$.
The sequence $\{0, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 1, \ldots\}$ is generated by the polynomial $g^*(x) = 1 + x^3 + x^4$. Determine the numerator polynomial $d(x)$.

$$y(x)d(x) = g(x)$$

$$y_0 + (y_0 g_{p-1} + y_1)x + (y_0 g_{p-2} + y_1 g_{p-1} + y_2)x^2 + \ldots = d_0 + d_1 x + d_2 x^2 + \ldots$$

$$d_0 = y_0$$

$$d_1 = y_0 g_{p-1} + y_1$$

$$d_2 = y_0 g_{p-2} + y_1 g_{p-1} + y_2$$

$g_0 = 1 \quad g_1 = 1 \quad g_2 = 0 \quad g_3 = 0$

$y_0 = 0 \quad y_1 = 1 \quad y_2 = 0, \; y_3 = 0$

$d_1 = 1 \quad d_i \text{ for } i \neq 1 = 0$

$\therefore d(x) = x$